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Self-diffusion of ethylammonium nitrate ionic liquid confined between modified polar glasses

Andrei Filippov [a,b,*], Oleg I. Gnezdilov [c,d], Alexander G. Luchkin [c], Oleg N. Antzutkin [a,e]

[a] Chemistry of Interfaces, Luleå University of Technology, SE-97187 Luleå, Sweden
[b] Kazan State Medical University, 420012 Kazan, Russia
[c] Kazan Federal University, 420008 Kazan, Russia
[d] Zavoisky Physical-Technical Institute, FRC Kazan Scientific Center, Russian Academy of Sciences, 420029 Kazan, Russia
[e] Department of Physics, Warwick University, Coventry CV4 7AL, UK

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ABSTRACT

Ethylammonium nitrate (EAN) ionic liquid confined between flat polar glass plates demonstrates variable diffusivity that is sensitive to an external static magnetic field. Outside the magnetic field, diffusivity between the plates is higher than that in the bulk. However, after placing the system in a strong static magnetic field, the diffusivity gradually decreased. These processes occur during transformations between phases formed in EAN subjected to micrometer-size restrictions outside and within the magnetic field (Filippov et al., *J. Mol. Liq.* [2018] 268, 49). In this study, we used samples of two types: (i) with roughened surface formed by treatment of the glass plates with aqueous solutions of hydrofluoric acid and (ii) with vacuum deposited TiO_2 layers with a thickness of ca. 1 μm at glass-plate edges. Neither the surface modification of the glass plates, nor the TiO_2 layers controlled thickness of EAN confined between glass-plates significantly changed the above-described effects, which have been observed in studies using untreated glass plates. Therefore, the range of systems with detected phase transformations in EAN and accompanying effects, such as accelerated diffusivity and change in diffusivity under the influence of a static magnetic field, was expanded to the systems with roughened surfaces and the systems with TiO_2 layers controlled inter-plates distances. Results of experiments with roughened surfaces additionally suggested that the phase transformation of confined EAN in the external magnetic field is isotropic in nature rather than a phase transition from “layered to bulk” structures.

1. Introduction

Ionic liquids (ILs) are salts prepared from organic cations and either organic or inorganic anions that remain in a liquid state near room temperature. Their design and related properties provide novel potential applications for this new class of material [1–3]. Alkylammonium nitrates have three readily exchangeable protons on the -NH_3 group of cations; therefore, they belong to a class of so-called “protic” ionic liquids [2]. Ethylammonium nitrate (EAN) is the most frequently reported protic IL [2,4], which is used as a reaction medium, as a precipitating agent for protein separation [5], and as an electrically conductive solvent in electrochemistry [6]. Like water, EAN has a three-dimensional hydrogen-bonding network and can be used as an amphiphilic self-assembly medium [7]. Confined ILs [8–12], particularly EAN [9,12–16], have attracted special interest during the last few years. Enhanced diffusion of ethylammonium (EA) cations has been observed for EAN confined between glass plates [12]. Magnetic field strength is an important external variable that may influence phase and dynamic properties of some liquid systems, such as liquid crystals with molecules having anisotropic magnetic susceptibility [17] and solid materials, such as organic-based semiconductors [18]. There are reports on magnetic ionic liquids (MILs), which contain ferromagnetic ions in their chemical structure, but no perceptible magnetic field effects have been observed for non-magnetic ionic liquids [19,20] until it was found recently that both self-diffusion and NMR relaxation of EAN and propylammonium nitrate (PAN) confined between glass plates with inter-plates distances of ca. 4 μm reversibly alter after sample placement in a strong static magnetic field [14,15]. The main factors responsible for this effect are the availability of protons in the protic EAN and the polarity of the surface, while the exchange rate of -NH_3 protons plays a crucial role in the observed processes [14]. It has been suggested that the processes influencing the dynamics of EAN in this confinement are the phase transformations of EAN [14,15]. It has also been demonstrated that the process can be described well by the Avrami equation, which is typical for autocatalytic processes [15]. The transition can be stopped by freezing the sample. Cooling and heating investigations showed differences

* Corresponding author at: Chemistry of Interfaces, Luleå University of Technology, SE-97187 Luleå, Sweden.
E-mail address: andrei.filippov@ltu.se (A. Filippov).

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in freezing and melting behaviour of the sample that had been exposed or not exposed to the static magnetic field [15]. For EAN-water mixtures confined between glass plates, the diffusivity of the ethylammonium cation decreases with increasing concentration of water, while the measured diffusion coefficient of water remains unchanged [16]. It has been suggested that a static magnetic field has the effect of gradually decomposing the EAN-water phase into a neat EAN liquid phase and a "solid-like" water phase adsorbed onto the surface [16].

Only systems with flat, polished glass surfaces separated by ~4 μm by confined ILs were used until now. Therefore, it remains unclear how the planarity of the glass surface, its specific area and inter-plates distances in the micrometer-scale range may influence the dynamics of the confined ionic liquid and the effect of the static magnetic field. In this work, we prepared and studied two types of samples. First one was prepared using roughened glass plates treated with hydrofluoric acid. The structure of the treated surface was analysed by atomic force microscopy and electron scanning microscopy. The second type of samples was prepared using flat glass plates with vacuum-deposited TiO_2 layers of ~1 μm thickness at two opposite edges of the glass plates to additionally control thickness of EAN layers confined between glass-plates. The diffusivity of EAN confined between glass plates and the effect of applying a strong static magnetic field were studied by ^{1}H NMR.

2. Materials and methods

2.1. Sample preparation

The chemical structures of ions of EAN are shown in Fig. 1. EAN was synthesised and characterised at Chemistry of Interfaces of Luleå University of Technology as described previously [12]. The quantity of water in the synthesised ILs was <0.055 wt% as determined by Karl-Fisher titration (Metrohm 917 Karl Fisher Coloumeter with HYDRANAL reagent).

Confined ILs were prepared with glass plates settled in a stack (see Fig. S1 in the Electronic Supporting Information, ESI). The plates measuring 14 × 2.5 × 0.1 mm (Thermo Scientific Menzel-Gläser, Menzel GmbH, Germany) were used. For experiments reported previously [12,14], the plates were thoroughly cleaned before sample preparation (see the ESI in [12,14–16]).

For experiments with treated glass surfaces, the glass plates were placed in 4 wt% hydrofluoric (HF) acid aqueous solutions for 40 min. Wet chemical etching of glasses in aqueous HF solutions is a subject that has studied for many years [21,22]. When a smooth surface is etched the surface roughens and cusp-like structures are formed. Its formation is related to the presence of flaws in the surface before etching. The size or depth of these flaws varies from microcracks present after polishing to larger cracks formed by mechanical action such as grinding or particle erosion. Closed flaw or microcrack is etched open and this surface defect is gradually transformed into a cusp as etching proceeds [22]. Untreated and HF treated glass plates were analysed by BET method [23]. Nitrogen adsorption-desorption data were recorded at 77 K on a Micromeritics ASAP 2010 instrument. Prior to measurements, the samples were outgassed at 300 K under vacuum for 10 h. BET surface area was 0.1 ± 0.01 m^2/g and 0.05 ± 0.01 m^2/g for the untreated and HF etched plates, respectively. These surface areas contain contributions mainly from microcracks (micropores with sizes < 2 nm), in which EAN cannot penetrate. At the same time, volume of the microcracks is extremely small, ~0.000011 cm^3/g.

AFM images of glass plate surfaces before and after the treatment are shown in Fig. 2A and B, respectively. For glass plates with deposited TiO_2 layers, reactive magnetron sputtering of titanium in an oxygen atmosphere was used. TiO_2 layers with a width of ca. 1.5 mm each were vacuum-deposited at two opposite edges of the glass plate (see Fig. 3). Thickness of the deposited TiO_2 layers was estimated by scanning electron microscopy (SEM). After preparation, the acid-treated and TiO_2-modified glass plates were additionally cleaned in the same manner as other (untreated) glass plates used in experiments.

A stack of glass plates filled with EAN was prepared in a glove box in a dry N_2 atmosphere. Samples were prepared by dropping 2 μL of the IL

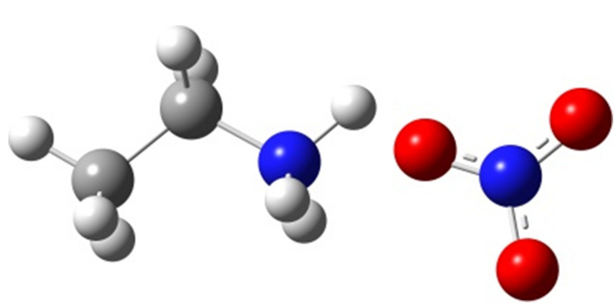

Fig. 1. Chemical structure of EAN, consisting of an ethylammonium (EA) cation (left) and a nitrate anion (right).

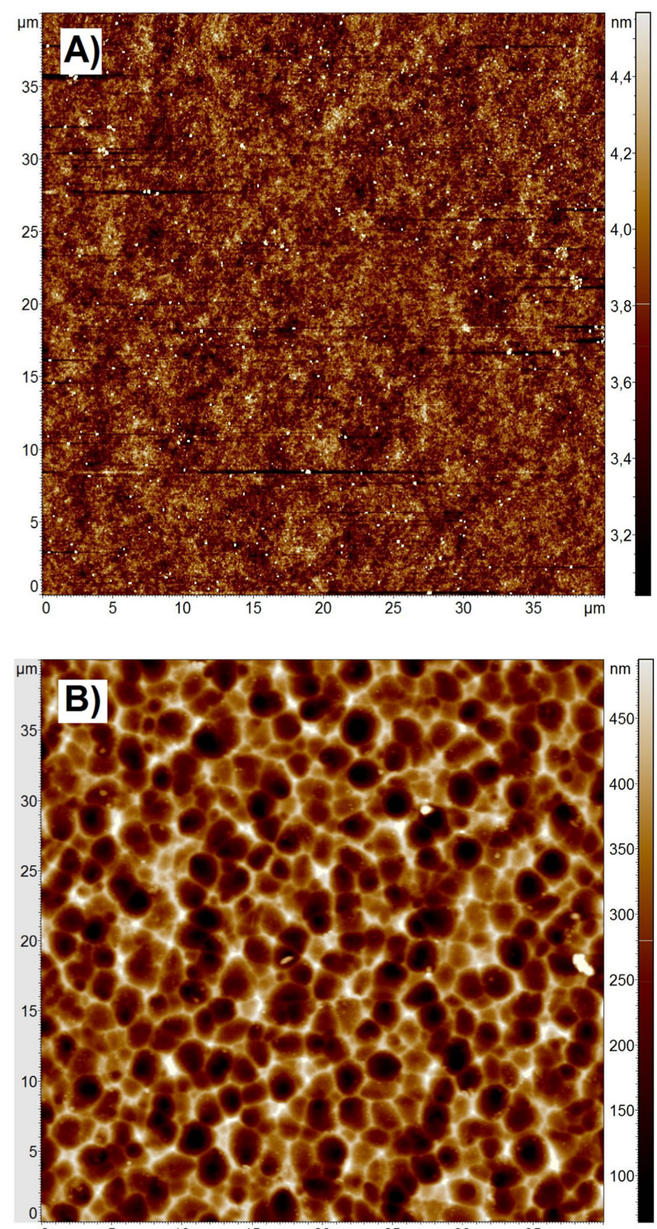

Fig. 2. Atomic force microscopy (AFM) images of untreated (A) and HF-treated (B) surfaces of glass plates.

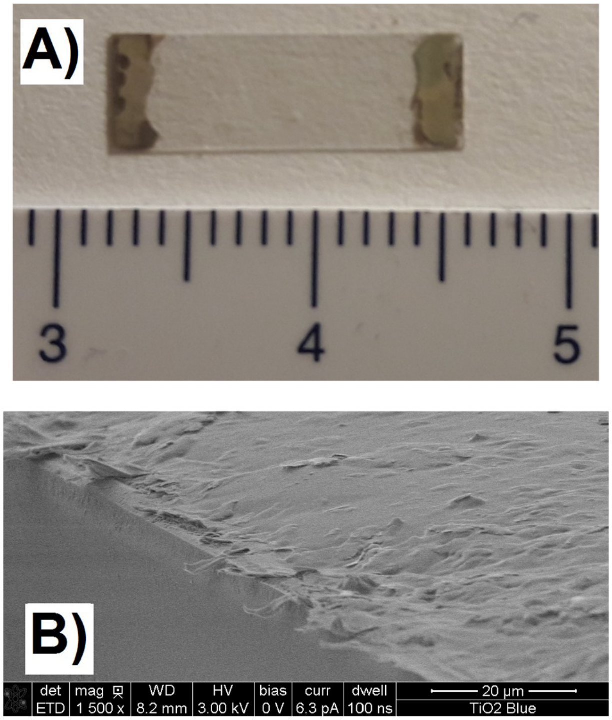

Fig. 3. (A) A glass plate (with dimensions 0.1 × 2.5 × 14 mm) with vacuum deposited TiO_2 layers of a thickness of ca. 1 μm at left and right edges of the glass-plate. The ruler below is scaled in cm units. (B) A SEM image near an edge of deposited TiO_2 on a glass-plate. Deposited TiO_2 is shown on the right side of the image as a light-grey flaky material, while a polished glass plate surface is shown as dark-grey on the left side of the image.

onto the first glass plate, placing another glass plate on top, adding 2 μL of the IL on top of this glass plate, etc. until the thickness of the stack reached 2.5 mm. IL that overflowed at the edges of the stack was removed by sponging. The sample consisting of a stack of glass plates was positioned in a rectangular glass tube and sealed. The mean spacing between the untreated glass plates was assessed by weighing the introduced IL, which yielded d ~ 3.8 ÷ 4.5 μm for the sample prepared with untreated EAN [12]. A detailed report of the sample preparation and characterisation has been described in our previous papers [12,14]. To allow equilibration of ILs inside the samples, experiments were started a week after the sample preparation.

2.2. Atomic force microscopy and scanning electron microscopy

Atomic force microscopy (AFM) images were recorded using NT-MDT NTEGRA AFM (Zelenograd, 124,460, Russia) in non-contact (tapping) mode using an NSG01 AFM tip. For SEM images, a Magellan 400 (FEI Company, Eindhoven, the Netherlands) electron microscope was used, with voltages in the range of 1 to 3 kV and currents of 6.3 to 25 pA.

2.3. Pulsed-field gradient NMR diffusometry

^{1}H NMR self-diffusion measurements were performed using a Bruker Ascend/Aeon WB 400 (Bruker BioSpin AG, Fällanden, Switzerland) NMR spectrometer with a working frequency of 400.27 MHz for ^{1}H (magnetic field strength of 9.4 T with magnetic field homogeneity better than $4.9 \cdot 10^{-7}$ T) using a Diff50 Pulsed-Field-Gradient (PFG) probe. Homogeneity of the magnetic field was estimated from the ^{1}H NMR line width. An NMR solenoid ^{1}H insert was used to macroscopically align the plates of the sample stack at 0 and 90° with respect to the direction of the external magnetic field (the same direction as that of the PFG).

The diffusional decays (DD) were recorded using the spin-echo (SE, at $t_d \le 5$ ms) or the stimulated echo (StE) pulse trains. For single-component diffusion, the form of the DD can be described as [24–26]:

$$A(\tau, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2}\right) \exp\left(-\gamma^2\delta^2 g^2 D t_d\right)$$

$$A(\tau, \tau_1, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2} - \frac{\tau_1}{T_1}\right) \exp\left(-\gamma^2\delta^2 g^2 D t_d\right) \quad (1b)$$

for the SE and StE, respectively. Here, A is the integral intensity of the NMR signal, τ and τ_1 are the time intervals in the pulse train; γ is the gyromagnetic ratio for protons; g and δ are the amplitude and the duration of the gradient pulse; $t_d = (\Delta - \delta/3)$ is the diffusion time; $\Delta = (\tau + \tau_1)$. D is the diffusion coefficient. In the measurements, the duration of the 90° pulse was 7 μs, δ was in the range of (0.5 ÷ 2) ms, τ was in the range of (3 ÷ 5) ms, and the amplitude of g was varied from 0.06 up to 29.73 $T \cdot m^{-1}$. The recycle delay during accumulation of signal transients was 3.5 s.

As has been shown in our previous work [12], restrictions (untreated glass plates) had no effect on the D of EAN in the direction along the plates, while for diffusion normal to the plates the effect of the plates on D was noticeable at t_d longer than 3 ms. For diffusion along the plates, D values were obtained by fitting Eqs. (1a) and (1b) to the experimental decays. All measurements were performed at 293 K (room temperature), therefore, we did not have a time gap until the thermal equilibration of the sample in the NMR probe. Measurements were started directly (in 1 ÷ 2 min) after placing the sample in the NMR probe, tuning and matching the probe. The time required for a single NMR diffusion experiment was around 30 s. Data were processed using Bruker Topspin 3.5 software. Eqs. (1a), (1b) and (2) and non-linear least squares regression were used to fit the experimental data to extract D values.

3. Results and discussion

3.1. Structures of untreated and treated glass plates

Representative AFM images of untreated and treated (with HF acid) glass plates are shown in Fig. 2A and B, respectively. More AFM and SEM images are presented in the ESI (Figs. S2–S5). The untreated surface is fairly flat, with variation in height of no >1 nm (Fig. 2A), while the treated glass surface shows variations in height of up to 400 nm (Fig. 2B) with inhomogeneity in disk type of sizes ~1 ÷ 2 μm and corresponding radii of curvature of ~0.5 ÷ 1 μm. Therefore, HF treatment leads to roughening of the glass surface, as it has been reported earlier [21,22]. Surface area of micropores decreases (BET data), while the surface area of macropores (larger than 50 nm) increases. As a first approach, we suggested the factor of increasing the surface area of untreated plate relative to an ideal flat surface as 1. This means that inhomogeneity of the surface does not increase the surface area. To estimate the factor of increasing the macroporous surface area for the case of an HF acid-treated surface, we took a model, in which the surface was modelled by a number of adjacent disks of a mean diameter of 2 μm and a height of 300 nm. The model is presented in the ESI, Fig. S6. Our calculation showed the factor to be ~1.8. This approach underestimated the factor, because many of the disks in Fig. 2B have a diameter of <2 μm. But this underestimation partly compensated for their lower height, which is on the order of 100 nm. For glass plates with deposited TiO_2 layers at the edges of glass plates, surface structure of TiO_2-uncovered areas is considered to be similar to that of untreated glass plates, as observed by SEM.

3.2. Diffusivity

3.2.1. Diffusivity in restricted geometry

Diffusion of ethylammonium cation in samples prepared with HF acid-treated glass plates was studied along the glass plates, as well as in the normal direction, in the same manner as has been studied before with untreated plates [12,14]. The dependences of the diffusion decays on the diffusion time were obtained during the first 10 min after placing the sample in the magnetic field of the spectrometer. For the measurements along the plates, the diffusion decays demonstrated single-component forms (Eqs. (1a) and (1b)) at any diffusion time in the range of 3 ÷ 1000 ms. For measurement of diffusion in the direction perpendicular to the surface of the plates, the diffusion decays demonstrated complicated forms varying with diffusion time, as shown in Fig. 4. Diffusion decays obtained for short diffusion times are subjected to the effects of collision of ions with solid surfaces to a minimal degree. It was found that diffusion coefficients obtained from the analysis of these diffusion decays were not dependent on the orientation of the glass plates relative to the static magnetic field. Therefore, diffusion dynamics of ions of EAN confined between polar glass plates is isotropic in nature.

To analyse the diffusion-time dependence of the diffusion decays, we used a known expression [27] that has been applied previously to analyse the diffusion decays of EAN between plane glass plates [12,14]. Planar restriction is one of regular geometries, in which diffusion of a confined liquid has been analytically resolved. The expression for the diffusion decays is presented in Eq. (2) [24,28]:

$$A(\delta,\Delta,g) = \frac{2[1-\cos(\gamma g\delta d)]}{(\gamma g\delta d)^2} + 4(\gamma g\delta d)^2 \times \sum_{n=1}^{\infty}\left\{\frac{1-(-1)^n\cos(\gamma g\delta d)}{\left[(n\pi)^2-(\gamma g\delta d)^2\right]^2}\times\exp\left(-\frac{n^2\pi^2D^*\Delta}{d^2}\right)\right\} \quad (2)$$

where D^* is the diffusion coefficient "unaffected" by collisions with the walls, and d is the distance between plates. This is an iterative process, which was applied to obtain the best fit for diffusion decays using Eq. (2). The fits of the whole set of experimental data of diffusion decays with different t_d were obtained after a number of iterations being varied up to 1000. The separation between planes was first estimated as in the case of plane plates [12,14], but then used as a fitting parameter, together with $D^* = 6.5\cdot10^{-11}$ m^2/s, to better match the experimental time-dependent diffusion decays. Eq. (2) and non-linear least squares regression were used to fit the whole experimental data set shown in Fig. 4. Our data does not demonstrate a "diffusion diffraction" effect, i.e., periodic oscillations in diffusion decays, which is expected for diffusion in regular structures [29] and in systems with a specific distribution of restrictions with a characteristic geometry [30]. We also tested fits using a number of distributions of d, such as Gaussian and log-Gaussian distributions to fit the experimental diffusion decays in Fig. 4 without identifying any satisfactory match. However, the best fits were obtained with an empirically chosen discrete distribution of d (Fig. 5, open circles). These best fits of calculated diffusion decays to the experimental ones are shown in Fig. 4 by solid lines; they describe the whole experimental set of diffusion decays satisfactorily well. The mean distance between planes in this simulation is 4.1 μm. The type of distribution of distances between plates is similar to that obtained for EAN confined between untreated flat glass plates [12,14] (see Fig. 5, solid circles), while the mean distance is slightly larger than that obtained for the untreated flat glass plates (3.8 μm).

Diffusion of ethylammonium cations in the sample prepared with TiO_2-treated edges of glass plates was also measured in the direction perpendicular to the surface of the plates. The diffusion decays demonstrated complicated forms varying with diffusion time, qualitatively very similar to those shown in Fig. 4. These decays together with their best fits to Eq. (2) are shown in Fig. S7 of the ESI. Discrete distribution of d for this case is shown in Fig. 5 (triangles). The mean distance between planes (4.9 μm) in this simulation was estimated from the obtained distribution of d that agrees well with the increased inter-planar distance relative to the sample with untreated glass plates.

3.2.2. Effects of confinement and magnetic field on diffusion

Just after placement of the sample in the spectrometer, the measured diffusion coefficient of the ethylammonium cations in the sample prepared with untreated glass plates is higher than that of EAN in the bulk [14–16]. A similar phenomenon is observed for confined EAN samples prepared using HF-treated glass plates and confined EAN samples with vacuum deposited TiO_2 at edges of glass plates (Fig. 6). Thus, the latter modifications of the system do not affect qualitatively the observed phenomenon, since the value of the diffusion coefficient is only ~12% higher than that in the EAN confined between untreated glass plates. During exposure to the static magnetic field, the diffusion coefficient of the ethylammonium cations in the sample prepared

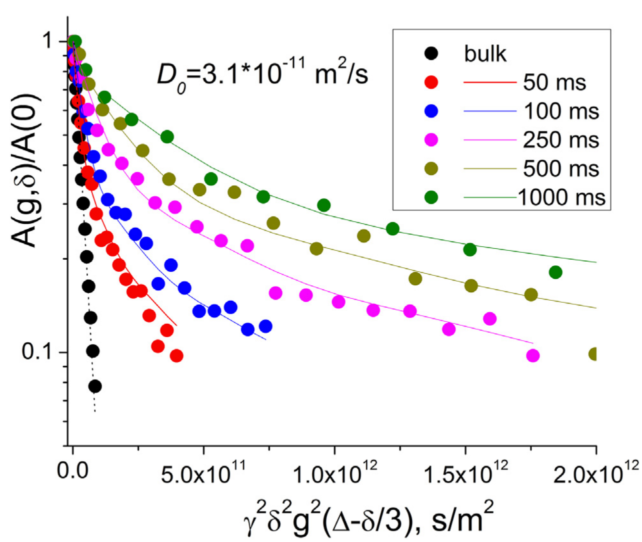

Fig. 4. The diffusion decays of the ^{1}H NMR signals of ethylammonium cations recorded at 293 K in bulk EAN (black circles) and in EAN films confined between parallel glass plates (HF-treated surfaces) with gradient directions perpendicular to the plates just after placing the sample in the magnetic field of the NMR spectrometer at different diffusion times, t_d (see data in the figure insert). The best fits of suitable expressions (Eq. (2)) for restricted diffusion are represented by solid lines corresponding to the data points. D_0 is the diffusion coefficient of ethylammonium cations in the bulk EAN.

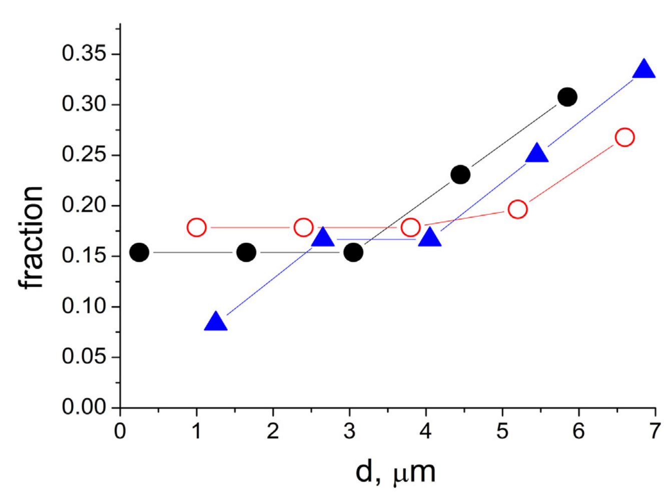

Fig. 5. Distance size distribution used to fit diffusion decays of ethylammonium cations for diffusion normal to the plates for untreated (solid circles) [14], HF acid-treated glass plates (open circles) and plates with vacuum deposited TiO_2 at edges of glass plates (triangles).

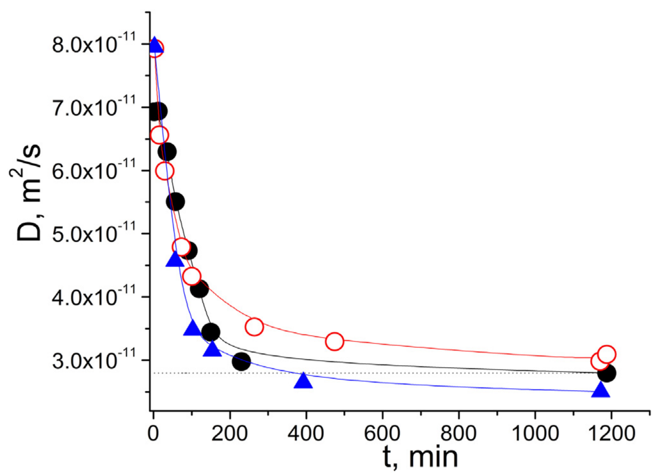

Fig. 6. Evolution of the diffusion coefficient of ethylammonium cations after placing EAN confined between polar glass plates in a magnetic field of 9.4 T for untreated glass plates (solid circles) [14], HF acid-treated glass plates (open circles) and glass plates with vacuum deposited TiO_2 at edges of glass plates (triangles). D was measured with gradients along the plates, $t_d = 50$ ms.

using untreated glass plates decreases as a function of time (Fig. 6, solid circles) [14–16]. Similar effects are observed for samples prepared for samples prepared with HF- and TiO_2- treated glass plates (see Fig. 6, open circles and triangles, respectively). Rates of a decrease of the diffusion coefficient of ethylammonium cations are comparable for all these three samples, at least in the initial part of the process (during first 50 min), but with clear deviations at longer times for samples prepared with treated glass plates to be discussed below.

In papers [14–16] observed effects were treated as phase transformations occurring in EAN under the influence of the polar surfaces of the glass plates and an external static magnetic field. It has been suggested that EAN, being confined, forms a phase that is different from the bulk. This phase is characterised by a higher diffusivity of both ethylammonium cations and nitrate anions. Under the influence of the magnetic field, this phase transforms into a new phase, different from both previous phases [15]. Thermodynamic proofs and kinetic dependences have been presented [15]. From our current study, it is evident that such effects are relevant not only to EAN systems prepared with plane glass surfaces of plates stacked on each other, but also to systems with quite rough surfaces, with a typical radius of curvature on the order of ~0.5–1 μm and for the stack of glass plates with vacuum deposited TiO_2 at edges of glass plates. Performed modifications of glass plates did not significantly change the observed effect. Therefore, we suppose that the observed phase transformations and accompanying effect, such as the enhanced diffusion and change in diffusion under the influence of the magnetic field, are related to a greater degree to the surface properties than to the geometry of the surface or inter-plate distances in the micrometer-scale range.

Dynamics of ionic liquids at interfaces with solid surfaces plays a key role in a range of applications [2,8–10,19,31]. Molecular dynamics simulations [8,9,19] and experiments with neutral and charged surfaces [8,9,19,31] demonstrated layering of ions (particularly, for EAN [9,19]) near solid surfaces, while other ions that remained in the central core were less affected by interactions with surfaces. The layering is very pronounced near the surfaces and it persists for over 2 nm with a long-range decay of the layered structure with a characteristic decay length of the order of 10 nm [9,19,31,32]. Thus, interactions of ILs with surfaces usually lead to transformations of isotropic “spongy”-like structures typical for bulk ILs to the layered structures in the nanometer-scale range. Our experiments with alkylammonium nitrates confined between flat glass plates demonstrated isotropic phase transformations occurring first without and later after application of the external magnetic field [14–16], which affects dynamics [12–16] and melting behaviour [15] of protic ionic liquids in confinement between polar glass plates. These phases are formed in the whole volume of confined EAN. In this our study we demonstrated that EAN confined between roughened glass plates and between plates with vacuum deposited TiO_2 layers at glass-plate edges (thus with a different distribution of inter-plate distances), also experiences phase transitions when exposed to the external magnetic field. There are two types of changes observed in Fig. 6 via the phase transformation curves for EAN samples confined between differently modified glass plates. (i) Initial values of diffusion coefficients are somewhat higher for samples with modified glass plates. Indeed, modifying of plates in both cases lead to a change in the distance distribution curves (Fig. 5) and averaged distances calculated from these curves to somewhat higher values. It is known that for larger pores retardation effects of pore walls for self-diffusion of confined liquids is smaller if the liquids maintain their structure [33]. The latter trends can be invoked to explain a slight increase of the initial diffusion coefficient of EAN in the samples with modified glass-plates (Fig. 6). (ii) During the exposure of the three samples in this study to the static magnetic field of 9.4 T for a long time (>1200 min), the diffusion coefficient of EA decreases following different kinetics and levels out to somewhat different values (different by ca. 5–10%). We anticipate that these differences between the named three samples are correlate with the fraction of small pores in each sample. From our preliminary (unpublished) data on diffusion of EAN confined in porous (Vycor) glasses with the pore size ranged between 40 and 100 nm, we observed that no phase transformations occurred under the influence of the external static magnetic field of 9.4 T in such small pores. Reason for the latter observations are yet not clear, but we can take this fact in consideration to understand the phenomenon investigated in this study.

4. Conclusions

We studied the diffusion of ethylammonium cations of EAN confined between roughened glass surfaces with increased surface area in the macroporous range formed by treatment with an aqueous solution of hydrofluoric acid and between flat glass plates with vacuum deposited TiO_2 layers at glass-plate edges. It was shown that the surface roughness of the glass plates, corresponding increase in the macroporous surface area and increase in the inter-planar distance do not change principally the effects observed in the case of the plane plates; these effects are related to phase transformations in the ionic liquid. Therefore, the observed transformations and accompanying effects, such as enhanced diffusion and change of diffusion under the influence of the magnetic field, are related to a greater degree to the ionic liquid - surface interactions than to the local geometry of the surface. This study also demonstrates that isotropic phase transformation mechanism rather than “bulk to layered” transformations occurring with EAN in confinement.

Assuming that the structure and dynamics of ionic liquids are significantly altered in confinement and in the magnetic field, we propose that the results will have strong implications for interface-intensive applications of ILs, such as their use in machinery, electrochemical and electro-magnetic systems.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2019.04.021.

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